LAB EXPERIMENT

3. Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to show outline using Canny function.

Code:

```
import cv2
import matplotlib.pyplot as plt

# Read the image
img = cv2.imread(r"C:\Users\Admin\Downloads\images.jpg")

# Check whether the image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Detect edges using Canny
    edges = cv2.Canny(gray, 100, 200)

    # Display original image
    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis("off")

    # Display outline image
    plt.subplot(1, 2, 2)
    plt.imshow(edges, cmap="gray")
    plt.title("Canny Edge Detection")
    plt.axis("off")

    plt.show()
```

Original Image



Canny Edge Detection

